

TRACK 3 • RESILIENT SYSTEMS • STANFORD CLEANTECH HACKATHON 2026

CREW AIR

A reading at the work site when the nearest monitor is 48 km away: the record a utility crew goes in with after a fire, and the same hours for buildings and the grid.

Rustam Salavatov • Asher Popper • live at airsignal.app/crew

THE RULE EXISTS.
THE DATA IS 48 KM AWAY.

40 km

the cell of the smoke model a dispatcher looks at; it reads low right next to a fire

48 km

to the nearest regulatory monitor at one of our test sites; the rule wants the air determined and logged before the crew goes in

8,507

Cal/OSHA heat-rule violations since 2019, $8.1M in penalties; smoke is the same sheet

Open-Meteo CAMS grid • AirNow monitor distance measured at Sacramento Canyon • DOL OSHA enforcement data, July 2019 to Sept 2026

HEAT GETS CITED. SMOKE DOES NOT.



8,507

heat violations since July 2019, $8.1M in penalties

19

smoke violations, 16 of them in 2021. None in 2020, the worst smoke year

DOL OSHA enforcement API (Cal/OSHA is a state plan), violations issued Jul 2019 to Sep 2026, deleted citations excluded • "days in smoke" = farmworker-days at AQI 151+, EPA AirData × BLS QCEW

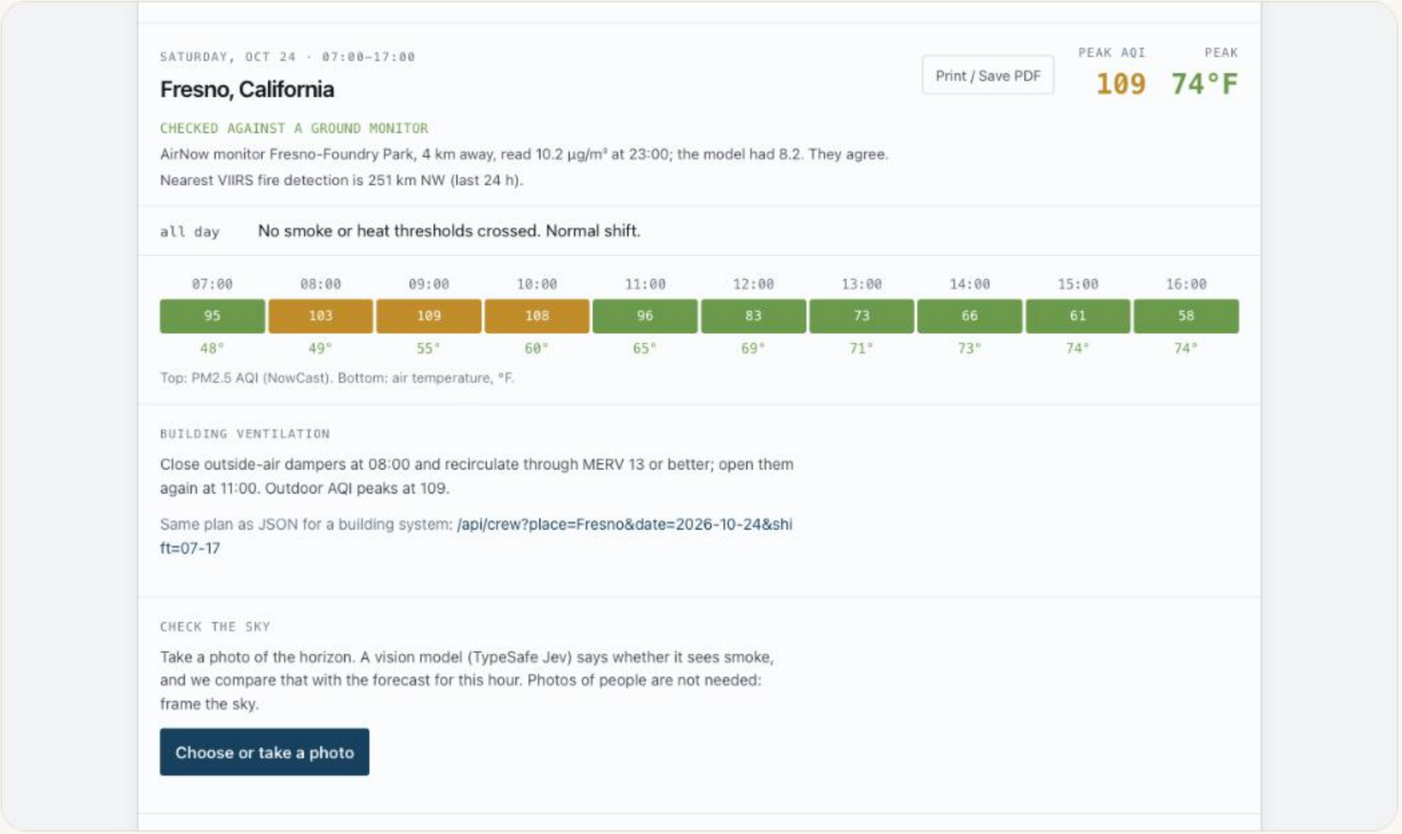
PLACE IN. SHIFT PLAN OUT.

INPUT

Site: Fresno

Shift: 7:00–17:00

Out: the hour each rule starts, the clean window for heavy work, when to close outside air, a sheet to sign.



Live at airsignal.app/crew • AQI is an EPA NowCast from hourly PM2.5, not the 24-hour mean forecast feeds give

EVERY PLAN IS CHECKED FOUR WAYS

Model: hourly smoke + heat forecast, NowCast

Satellite: NASA VIIRS fires within 50 km

Ground: AirNow monitors within 25 km

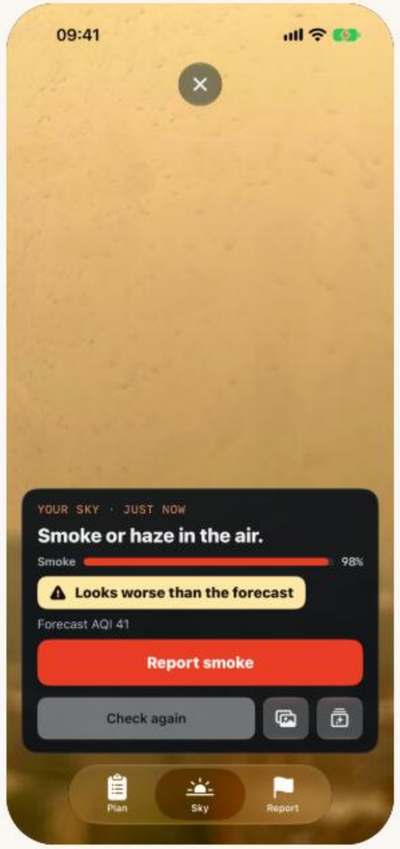
Crew: reports from the people on site



IN THE CREW'S POCKET



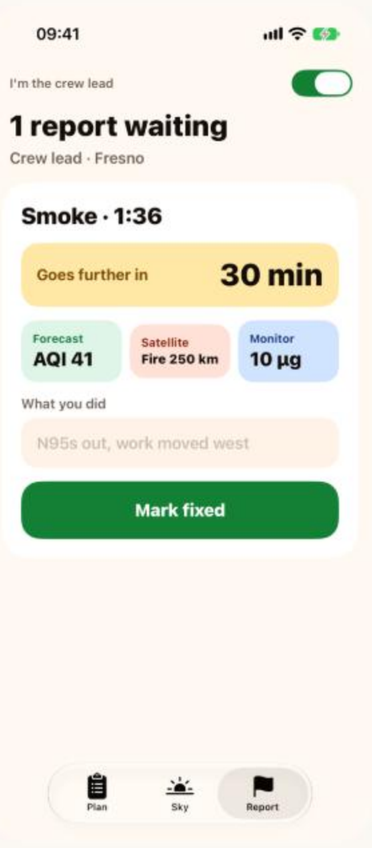
Plan + sensor task



Check the sky



Report

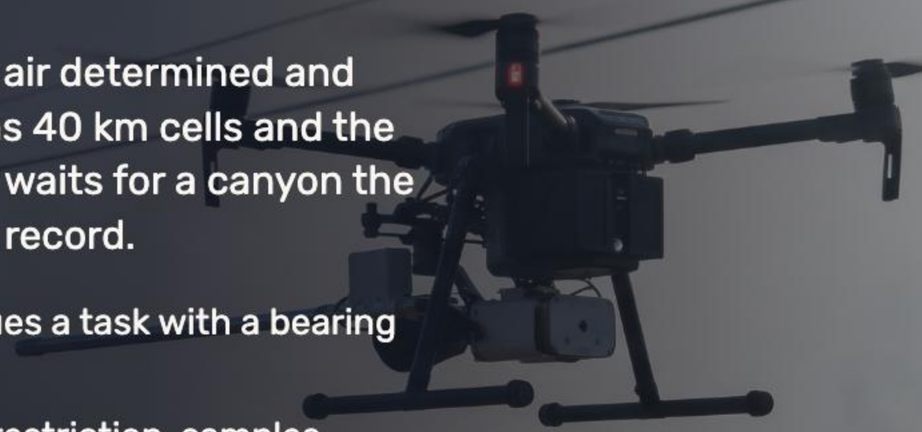


Crew lead

Native iOS, English and Spanish • the sensor task and the AQI record on the Plan tab • sky check on TypeSafe Jev: smoke call right 11–12 of 12 known-answer photos, never wrong • faces blacked out on the phone

LINE CREWS AFTER A FIRE: A READING, NOT A WAIT

A line is down behind a fire front. The rule wants the air determined and logged before the crew goes in; the smoke model has 40 km cells and the nearest monitor can be 48 km away. The dispatcher waits for a canyon the wind has already cleared, or sends people without a record.



1 • Trigger VIIRS sees the fire within 50 km; the API issues a task with a bearing and a distance (`dispatch`).

2 • Reading A sensor node, or a drone outside the flight restriction, samples PM2.5 at the work site and reports it; a photo of the sky is the second check.

3 • Record The plan prints with the reading, the hour, the rule and the N95 line, signed at the briefing: the document the crew goes in with, and the one an inspector asks for.

Who pays: the employer of the crew, per site, for the reading and the record; the utility is the channel and the pilot • Hours saved per outage is the pilot's number, not ours yet • §5141.1 requires N95s at AQI 151+, respirators above 500; it does not forbid the work • Drone payload: a concept, the API side is live

THE API SENDS FOR A READING

Smoke models have 40 km cells and read low next to a fire. At one of our test sites the nearest monitor is 48.7 km away.

So when a satellite sees fire within 50 km, the API now issues a task: a bearing, a distance, and “sample PM2.5 toward it”. A fixed sensor node answers; where there is no ground, a drone does, outside the fire’s flight restriction.

The reading comes back through the same report endpoint, and the next plan is corrected by it the way it is by an AirNow monitor today. Scout payload, a concept: an SPS30-class PM2.5 module and an LTE modem on a standard drone, intake ahead of the downwash; the API side is live, the payload is not built.

Honest limits: no flight over a fire (FAA TFR), autonomous flight needs a Part 107 waiver, and nobody needs “every second” sampling. Smoke takes 1–3 hours to travel 20 km; one reading every 15 minutes is what a plan needs.

```
GET /api/crew?place=36.95,-120.15
// near Madera, 24 Oct 14:30
"dispatch": {
  "sensor": "launch",
  "bearing_deg": 333,
  "distance_km": 9,
  "detections_50km": 1,
  "reason": "VIIRS fire 9 km NW: sample
    PM2.5 toward it, outside the
    fire's TFR, and report"
}
```

FREE FOR CREWS. PAID BY EMPLOYERS.

Open

Engine, rulebooks and the worker app, AGPL. Workers get it through unions and worker centers.

Paid

Per site: Spanish and English alerts, many sites, the log archive an inspector or insurer asks for, BMS feed.

Public data

Exposure and report counts by county, anonymized, published openly for researchers and agencies.

TRY IT TODAY

airsignal.app/crew

- **Intros to a union, legal aid or worker center**
- **One farm or crew, and one utility contractor, to pilot next summer**

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Scan: LinkedIn

THE LAW ALREADY SAYS WHAT TO DO

AQI 151

N95s for the crew, tell them, cut exposure

§5141.1, wildfire smoke

80°F

Shade up, water at hand

§3395, heat illness

95°F

High-heat procedures, watch the crew

§3395, heat illness

Every outdoor employer in California. Checking it is a person opening airnow.gov at 7 am.

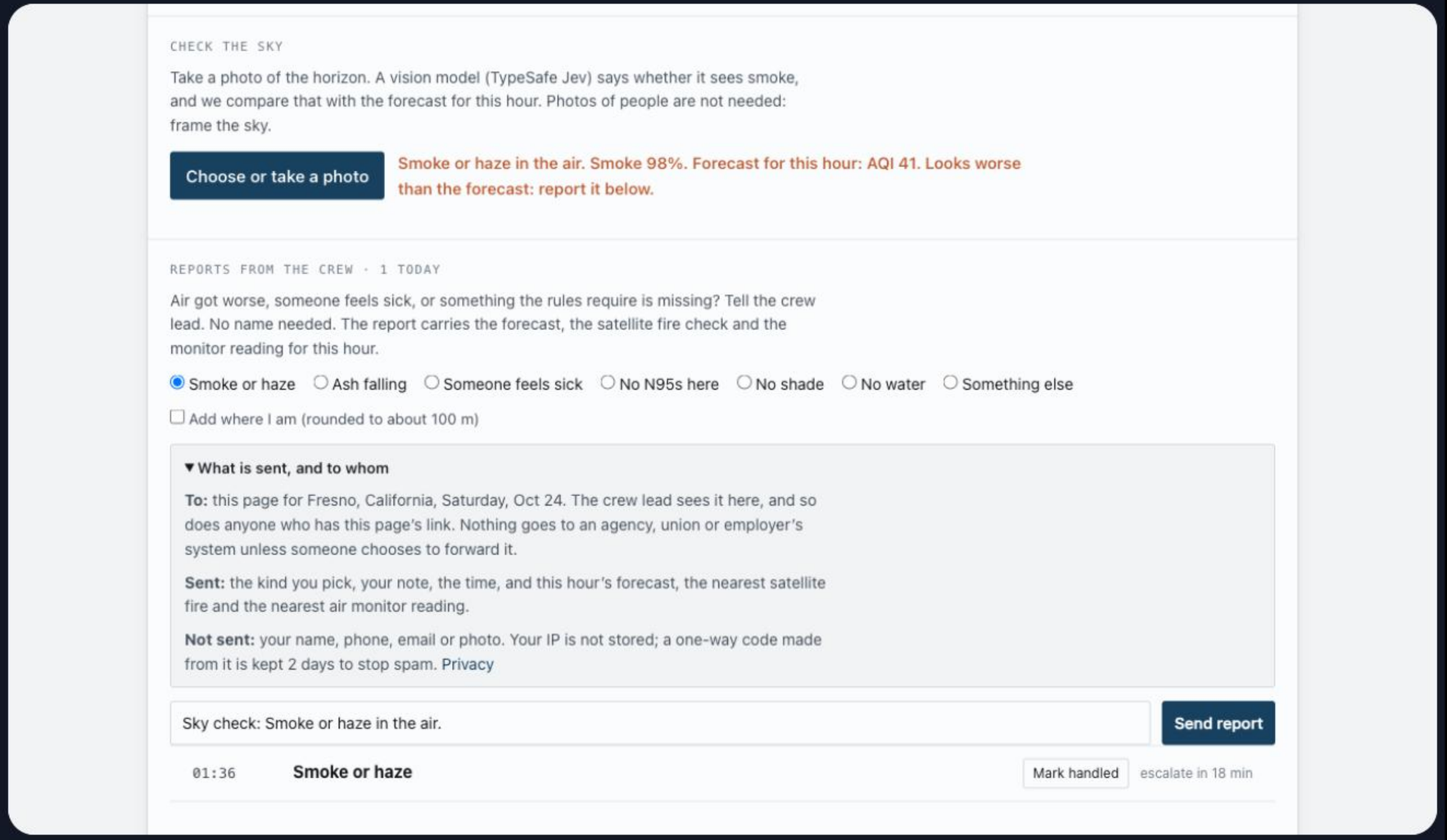
LIVE TODAY

No app needed: the plan, a map of where the smoke goes, the sky check and reports on the web. For a utility: 13 cities on one map.

airsignal.app/crew

airsignal.app/crew/grid • API: airsignal.app/api

Code: github.com/fortunto2/stanford-cleantech-2026



A REPORT THAT CAN'T BE IGNORED

1 • WORKER

"Smoke. No N95s here."

No name. The report carries the NowCast, the satellite fire and the monitor reading.

2 • CREW LEAD

Fixes it, marks it handled

§5141.1 requires employers to encourage these reports. This is that channel.

3 • AFTER 30 MIN

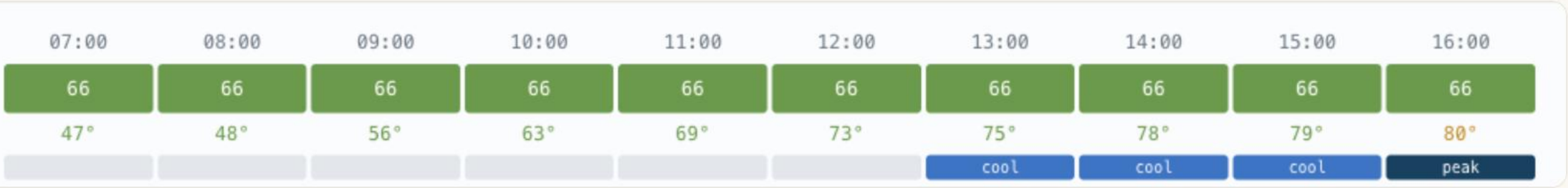
Goes further, anonymously

A Cal/OSHA complaint drafted with the evidence attached; the worker files it. Later: unions, legal aid.

All three steps live at airsignal.app/crew • we draft, never file for the worker • unions and legal aid need to agree to receive reports • retaliation is illegal, Labor Code §6310

THE SAME HOURS, FOR THE GRID

Heat and smoke days are the grid's worst days too. The peak price runs 4–9 pm every day; the house must be cooled before it, and in smoke it stays shut, so the air conditioner runs longer.



CUSTOMER

One strip: when it is hot, when it is smoky, when it is expensive. "Cool" marks the cheap hours to pre-cool; "peak" the PG&E E-TOU-C window that CCAs like SVCE bill on.

GRID

The same day as JSON (`home`, `alerts`, `plan.bms`) for a home agent or a DERMS. In Sept 2020 smoke cut CAISO solar ~30% below July and hour-ahead forecasts missed it by 10–50%: the smoke hour is a signal the grid does not have.

Rules, not an optimisation: no dollars or megawatts claimed, that needs a load model and a pilot • Solar: EIA Today in Energy, 30 Sep 2020; Juliano et al., ERL 2022

WHAT WE SELL

THE RULE, DONE ON TIME.

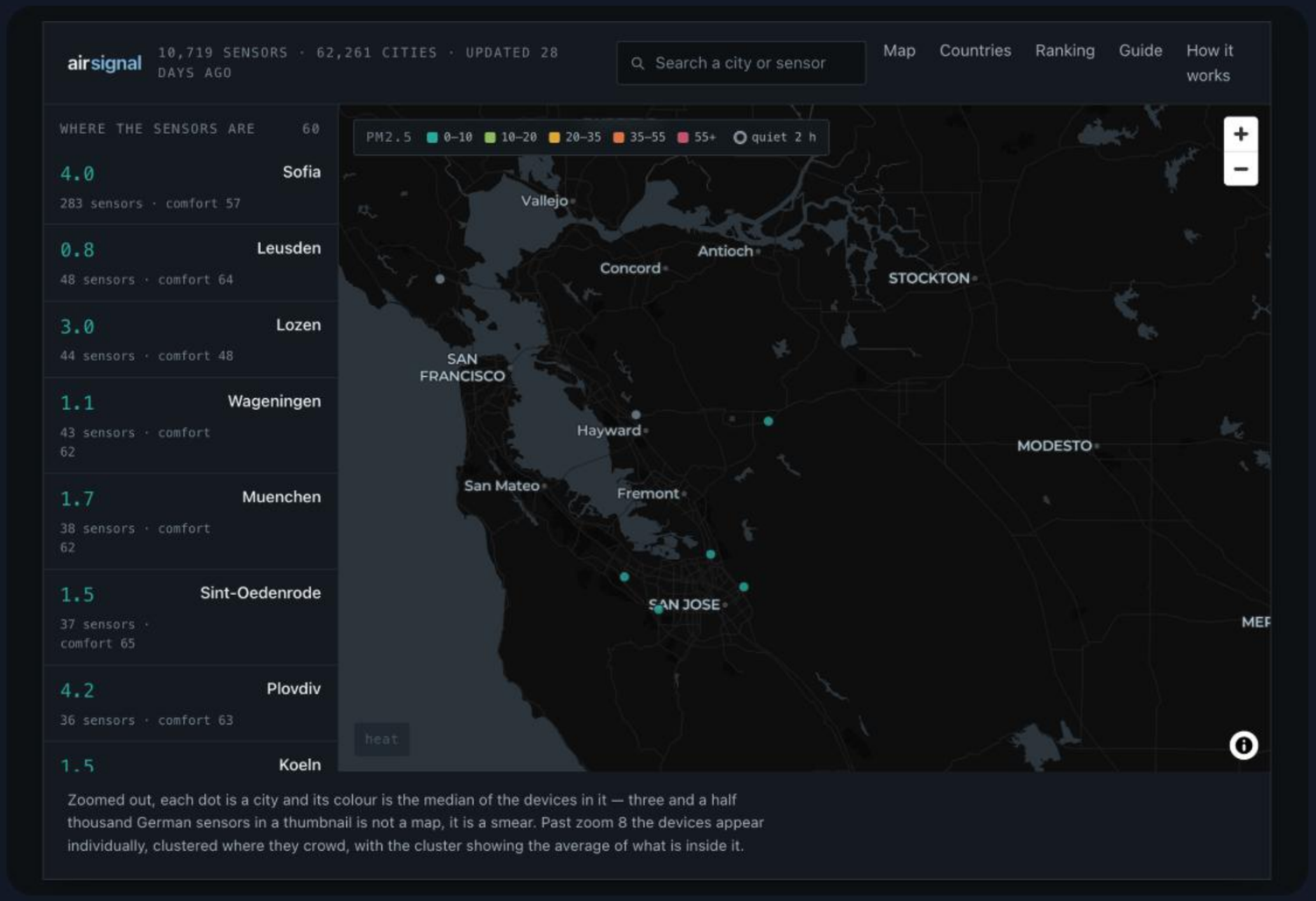
N95s on site before 151. Heavy work moved to the clean hours. Every crew report handled within 30 minutes. A sheet that proves it.

WHY ME

Back home in Sterlitamak, Russia, we got residents to report bad air with evidence. The agencies started to act.

Then I built Air Signal, a live air map: 10,719 sensors, 62,261 cities.

That's how I know sensors alone don't cover a job site: look at the Bay Area.



BUILDINGS: ACT BEFORE THE SMOKE ARRIVES

TODAY

BMS reacts

Indoor sensor turns red, then HVAC switches. People already breathed it.

WITH CREW AIR

BMS gets a forecast

Smoke due at 14:00 → at 12:00 recirculate, boost HEPA, close outside air.

Same engine, third rulebook: crews by law, homes by vulnerability, buildings by BMS.

WHO DOES WHAT TODAY

Product	Smoke data	Says what to do	Knows the law	Shift plan + log	Crew reports	Building
AirNow, Fire & Smoke Map	Now, by area	Generic advice	No	No	No	No
PurpleAir, IQAir	Now, sensors	Generic advice	No	No	No	No
Watch Duty	Fire alerts	Evacuation info	No	No	No	No
County SMS alerts (Ventura)	Threshold text	At 151 and 500	Smoke	No	No	No
OSHA Heat app	Heat only	Heat tips	Heat, federal	No	No	No
BMS outdoor-air add-ons	Outdoor AQ feed	Engineer writes rules	No	No	No	Yes
Crew Air	**Hourly NowCast, checked 4 ways**	**Yes, by the hour**	**Smoke + heat, as data**	**Yes**	**Yes, the ladder**	**Signal to BMS**

Checked 23–24 Oct 2026 from public product pages

WHAT WE HEARD

“My weather app already warns me.”

It warns **you**, **now**, with a 24-hour average. We tell the person responsible what to do **tomorrow**, by the hour.

“Sensors are too sparse.”

The model covers every site. Satellites, monitors and the crew check it.

“Smoke fines are rare. Why pay?”

Heat is: 8,507 Cal/OSHA violations since 2019, $8.1M. The usual miss: no plan on site. We print it.

“Overstory, Watch Duty already do wildfire.”

They work before and during the fire, for the utility: trees, fuel, perimeter. We work the hour the air changes, for the crew under the rule. A channel, not a rival.

ONE ENGINE, MANY RULEBOOKS

Farms first
Heat every summer, smoke in fire years

Construction
Same two rules, crews on site

Warehouses
Indoor heat rule §3396, since 2024

Schools, youth sports
Recess and practice: go, move or cancel

Oregon, Washington
Their own smoke rules: a new file

Spain, Italy
Heatwave limits on outdoor work